CSA0712 - Computer Networks

CO1-AT1 Short answers

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1. a, Role of Each OSI Layer.

OSI Layer	Role in Email Communication
Application Layer	User Composes and sends the email using SMTP
Presentation Layer	Convert data format, encrypts and Compresses the email if needed.
session Layer	Establishes, manages and terminates Communication session
Transport layer	Breaks data into segments, ensures reliable delivery using TCP.
Network layer	Adds IP addresses and routes packets to destination
Data link layer	adds MAC addresses, creates frames and detects errors.
Physical layer	Transmits bits over the network through Cables or wireless signals

b, Data Encapsulation Process:-

Application Data [Email Message]

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Presentation [Encryption]

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session [information added]

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Transport [TCP header] → Segment

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Network: IP header added → Packet

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Data link: MAC header & Trailer added → frame

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Physical: Frame Converted into Bits and transmitted

c; De-encapsulation Process:-

At the receiver, the data moves from Layer 1 to Layer 7

1. Physical : Receives bits
2. Data link : Removes frame header and trailer
3. Network : Removes IP header
4. Transport : Removes TCP header and reassembles segments
5. session : Maintains and closes the session.
6. Presentation : Decrypts and Converts data to original format
7. Application : Displays the emails to recipient.

Flow:

Bits → Frame → Packet → Segment → Data → original Message.

3. a, Roles of the Network Layer and Data link layer.

- Network layer : Determines the best path and forwards packets using IP address
- Data link Layer : Transfers data between directly connected devices using MAC addresses and detects transmission errors.

b, IP addressing and MAC addressing

- IP address : Identifies the source and desination across different network
- MAC address : Identifies devices within the same local network and changes at each hop.

c, Routing Decisions at Intermediate Devices.

- Routers examine the destination IP address
- They Consult the routing table to select the best path.
- The packet is forwarded to the next hop until it reaches the destination.

5. a, OSI Layers in Web Communication :-

- Application Layer : sends and receives the HTTP request / response.
- Presentation Layer : Formats and encrypts data if needed.
- session layer : Establishes and manages the Communication session.
- Transport layer : Breaks data into segments using TCP
- Network layer : Adds IP addresses and creates packets
- Data link layer : Adds MAC addresses and Creates frames
- Physical layer : Transmits bits over the network.

5, b. Flow of Data Units:-

Application Data → Segment (transport) → Packet (Network) → frame (Data link) → Bits (Physical)

c, Server Response to client :-

- The server processes the HTTP request and prepares the webpage.
- The response is sent back through the OSI layers in reverse order.
- The client browser receives the data, reassembles it and displays the webpage.

7. a, OSI Layers and Networking Devices.

- Switches work at Data link Layer using MAC addresses.
- Routers work at the Network Layer using IP addresses.
- The OSI Layers work together to transmit data efficiently.

b, flow of Data :-

Sender : Data → Segment → Packet → frame → Bits

Receiver : Bits → frame → Packet → Segment → Data.

c, Addressing for Correct Delivery :-

- IP addresses identify the source and destination across different networks
- MAC addresses identify devices within the local network.
- Routers use IP addresses to forward packets, while switches use MAC addresses to deliver data to the correct device.

8. a, OSI Layers in Real-time Voice (VOIP)

- Application Layer : Handles Voice Communication
- Transport Layer : Delivers Voice data quickly using UDP.
- Network layer : Routes Packets using IP addresses.
- Data link & physical Layers : Transmit frames and bits over the network.

b, VOIP vs file transfer :-

- VOIP : Requires low delay and Continuous transmission.
- File transfer : Prioritizes accuracy and reliability, even if it takes more time.

c, Layer Coordination :-

- Each OSI Layer performs its function and passes data to the next layer.
- Routers and switches forward the packets Continuously
- This Coordination ensures smooth, real time voice Communication with minimal delay.